

School ERP System

Executive Summary

The School ERP System is a comprehensive, unified digital platform designed to automate and streamline the complete academic, administrative, financial, and operational lifecycle of an educational institution. It connects students, parents, teachers, and administrative staff through a single trusted source of truth, eliminating data silos and reducing manual, paper-based processes across every department. The solution covers core modules including academic management, attendance, examinations, fees, HR, library, transport, communication, and real-time GPS-based student safety tracking. Role-based access control ensures every stakeholder sees only the features and data relevant to their responsibilities, while integrated reporting and analytics provide actionable insights for school management. By consolidating admissions, attendance, assessments, finance, transport, and parent communication into one cohesive system, the School ERP System improves operational efficiency, data accuracy, transparency, regulatory readiness, and overall institutional decision-making for a modern, future-ready learning environment. This document defines the product vision, team responsibilities, and key delivery milestones.

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1. Executive Summary

The School ERP is a unified, modular platform that automates the complete academic, administrative, financial, and operational lifecycle of an educational institution. It brings students, parents, teachers, and management onto a single trusted system, and extends into real-time student safety through an integrated Live GPS Tracking module.

This document consolidates the product vision, panel-wise responsibilities, functional scope, cross-panel connectivity, technology stack, and a detailed gap analysis of items that must be added to deliver a production-ready, client-grade release.

Core Pillars

Pillar	Description
Academics	Curriculum, classes, exams, results, report cards.
Finance & Fees	Fee structures, invoicing, online payments, concessions.
Administration	Admissions, staff, front-office, communication.
Communication	Announcements, SMS/Email/Push, parent–teacher chat.
Reports & Analytics	Operational, academic, financial and safety dashboards.
Live GPS & Safety	Real-time tracking, geofencing, SOS, trip history.

Business Objectives

- Automate academic and administrative processes end to end.
- Ensure data accuracy, consistency, and security across all modules.
- Improve communication among students, parents, and staff.
- Provide real-time insights through reports and dashboards.
- Give every parent live, verifiable visibility into their child's location and travel status.
- Reduce manual, paper-based effort and operational overhead.

2. Panels & Role-Based Access

The platform ships with the following panels. Each panel has a dedicated UI, permission boundary, and workflow. All panels are governed by centralised Role-Based Access Control (RBAC) configured from the Admin Panel.

2.1 Admin Panel

Super-user control plane for the institution.

- Full, unrestricted access to every module including configuration and system-wide settings.
- Creates and manages all other user accounts, roles, and permission levels (RBAC).
- Configures institution profile, academic year, sessions, grading scales, and fee heads.
- Configures and monitors the Live GPS Tracking module — geofence boundaries, alert thresholds, retention policy.
- Views system-wide audit logs, security settings, backup and restore controls.
- Manages integrations (SMS gateway, payment gateway, email, maps API, biometric/RFID).
- Final authority over academic year setup, fee structures, and institution-wide policies.

2.2 Administration Panel

Day-to-day operational management of the institution.

- Handles admissions, enrolment, class/section allocation, and document verification.
- Manages staff records, ID cards, and front-office activities (enquiries, visitors, gate pass).
- Oversees attendance, transport allocation, and communication with parents and staff.
- Reviews GPS & Safety reports (trip logs, geofence events, SOS incidents) for follow-up.
- Coordinates between departments to ensure smooth daily functioning.

2.3 Teacher Panel

Classroom, academic and communication workspace for teaching staff.

- Manages assigned classes, subjects, curriculum, and lesson plans.
- Marks daily attendance and records student behaviour and performance.
- Creates exams, evaluates answer sheets, publishes results and report cards.
- Uploads assignments, learning material, and question banks.
- Communicates directly with parents regarding student progress.

2.4 Student Panel

Self-service portal and mobile app for students.

- Accesses academic content, timetable, and academic calendar.
- Submits assignments and views examination results, grades, and report cards.
- Views personal attendance records and library issue/return history.
- Receives announcements and notifications relevant to class and exams.

2.5 Parent App

Trust layer between the institution and the family.

- Real-time view of the child's location on an interactive map (from bus to campus and back).
- Push notifications for geofence entry/exit, bus boarding/alighting, and SOS events.
- Access to attendance, results, homework, fee dues, and school announcements.
- One-tap chat with class teacher and quick payment of fees.
- Trip history for the last 30 days (configurable) and route ETA to the child's stop.

2.6 Accountant Panel

Full finance and billing control.

- Defines fee structures, generates invoices, tracks due/overdue payments.
- Processes online and offline payments, refunds, and concessions/scholarships.
- Maintains financial ledgers and generates collection and transaction reports.
- Manages transport fee billing linked to the Transport & GPS module.

2.7 Librarian Panel

Library resources and lending lifecycle.

- Book cataloguing, classification, and inventory management.
- Book issue, return, renewal for students and staff.
- Overdue tracking and fine calculation.
- Library usage and inventory reports.

2.8 Bus Driver App

Dedicated mobile app for drivers and route staff.

- Shows assigned route, stops, and estimated arrival times.
- Marks student attendance on boarding and de-boarding the bus.
- Automatically transmits live GPS location every 10–15 seconds.
- One-tap SOS emergency button that alerts admin and parents with live location.
- Receives alerts for route deviation and speed-limit violations.

2.9 HR Panel

Employee lifecycle for teaching and non-teaching staff.

- Employee profiles, departments, and designations.
- Attendance and leave management (with approval workflows).
- Payroll and salary processing, payslip generation.
- Performance appraisal and document repository.

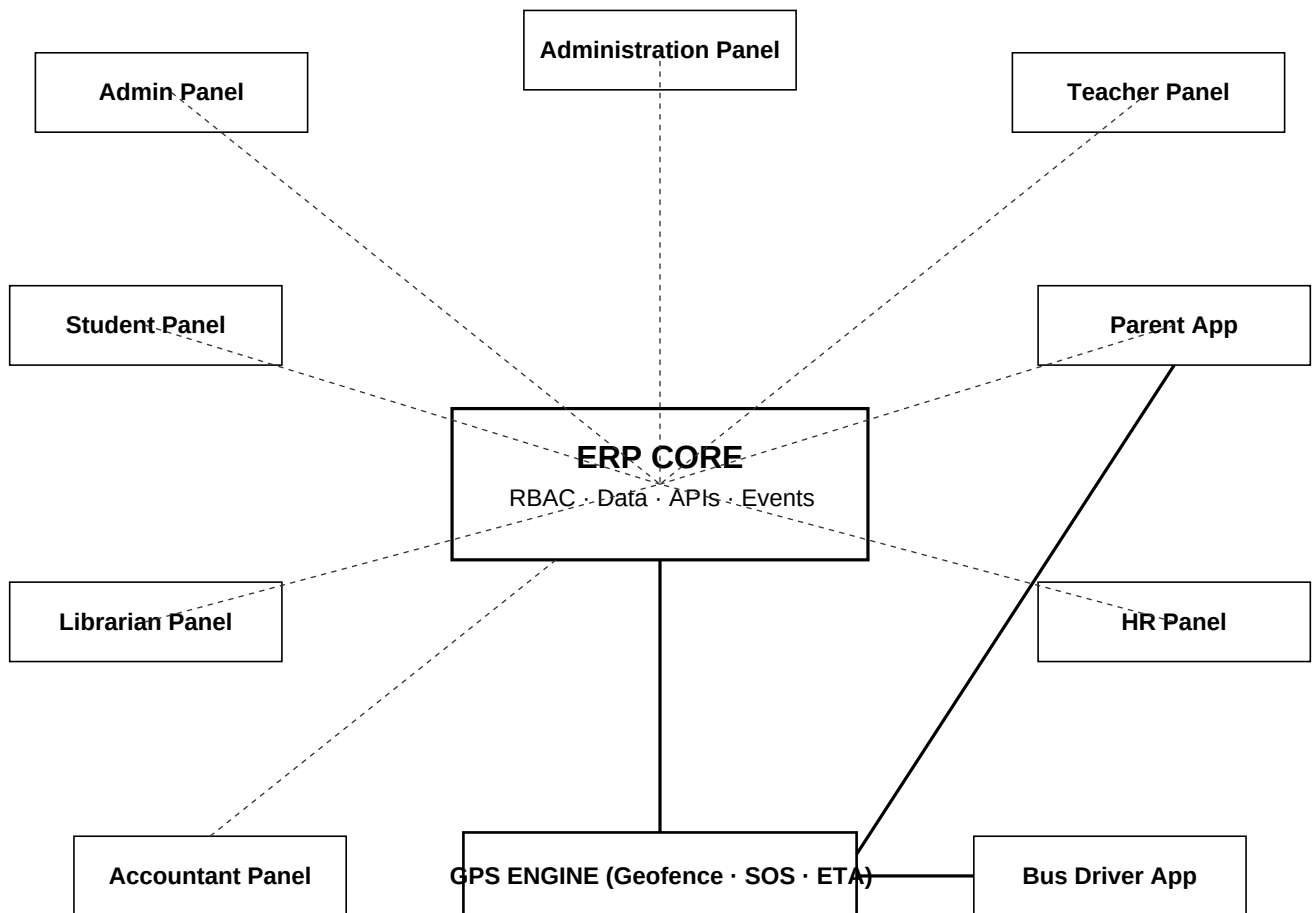
3. System Modules

Modules are independent yet integrated. Data flows automatically across them — for example, attendance captured by the GPS module at the school gate updates the Attendance module in real time.

#	Module	Purpose
1	Academic Management	Courses, subjects, classes, sections, syllabus, calendar.
2	Student Management	Admissions, profiles, promotions, transfers, documents.
3	Attendance Management	Manual, biometric, RFID and GPS-based attendance.
4	Examination Management	Exams, seating, evaluation, results, GPA.
5	Fees Management	Structures, invoices, online payments, concessions.
6	HR Management	Staff, leave, payroll, appraisal.
7	Library Management	Catalogue, issue/return, fines, reports.
8	Transport Management	Vehicles, routes, drivers, allocation, billing.
9	Inventory Management	Assets, stationery, uniforms, procurement.
10	Communication Management	Announcements, SMS/Email/Push, parent chat.
11	Reports & Analytics	Operational, academic, financial, safety dashboards.
12	Live GPS & Parent Safety	Live tracking, geofencing, SOS, trip history.
13	System Settings	RBAC, integrations, audit, backups.

4. Panel Connectivity & Workflow

The diagram below shows how each panel connects to the ERP core and where direct cross-panel data flows exist (for example, Bus Driver App → GPS Engine → Parent App).



Solid line: event-driven flow (GPS, SOS, notifications) | Dashed line: standard read/write via ERP Core

Reading the diagram: every panel authenticates against and reads/writes through the ERP core. Solid lines represent event-driven data flows that must remain low-latency — GPS updates, SOS alerts, teacher-to-parent notifications, fee receipts, and library dues.

5. Functional Requirements

5.1 Academic Management

- Manage courses, subjects, classes, and sections.
- Define curriculum, syllabus, and lesson plans.
- Assign teachers to classes and subjects.
- Academic calendar with holidays, events, and exam schedules.

5.2 Student Management

- Student admission, enrolment, and profile maintenance.
- Document repository (birth certificate, transfer, ID proofs).
- Track attendance, behaviour, and performance.
- Student promotion, section change, and transfer certificate generation.

5.3 Attendance Management

- Attendance by teacher, admin, biometric, RFID, or GPS gate event.
- Configurable rules (grace time, half-day, late-mark).
- Automatic parent notification on absence.
- Daily, monthly, and consolidated attendance reports.

5.4 Examination Management

- Exam creation, schedules, and seating plans.
- Question paper management and evaluation workflow.
- Result publication, report cards, and grade configuration.
- GPA / percentage / CCE grading support.

5.5 Fees Management

- Fee heads, structures, and installment plans.
- Invoice generation and automated due reminders (SMS/Email/Push).
- Online payment (UPI, cards, netbanking) and offline receipts.
- Concessions, scholarships, and refunds with approval workflow.

5.6 Communication Management

- Announcements at institution, class, and section level.
- SMS, Email, in-app messaging, and push notifications.
- Parent–teacher chat with read receipts and moderation.
- Event management with RSVP and reminders.

5.7 Library Management

- Book cataloguing (ISBN import), classification, and stock.
- Issue, return, renew, and reservation.

- Fines for overdue and lost books.
- Library usage analytics.

5.8 Transport Management

- Vehicles, routes, drivers, stops, and student allocation.
- Transport attendance and monthly billing.
- Powered end-to-end by the Live GPS Tracking module.

5.9 HR Management

- Employee profiles, departments, designations.
- Attendance, leave, and shift management.
- Payroll, tax deductions, and payslip generation.
- Performance appraisal cycles.

6. Live GPS Tracking & Parent Safety

Every parent can view their child's live location on a map — from the moment they board the school bus, throughout the journey, until arrival at school and back home — directly from the Parent App.

6.1 Key Capabilities

Capability	Description
Live Location	GPS trackers on buses and student ID cards/wearables update every 10–15 seconds.
SOS Emergency	One-tap SOS from driver/student app alerts admin and parents with live location.
Geofencing	Auto-notification on school/home geofence entry/exit and bus board/alight.
Trip History	30-day travel history accessible to parents and transport admin.
Route & ETA	Live route on map with estimated arrival at the child's stop.
Speed & Route Alerts	Alerts when a vehicle deviates from its route or exceeds a safe speed limit.

6.2 How It Works

GPS device on bus / ID card → Tracking server ingests every 10–15 s → Geofence & route engine evaluates events → Push notifications to Parent App → Live map view.

6.3 Privacy & Consent

- Location visible only to the verified parent/guardian and authorised school admin.
- Explicit parental consent captured during onboarding, with transparent disclosure.
- Location history auto-purged after a configurable retention window (default 30 days).
- Full audit log for every location view and SOS event.

7. Reporting Modules

Report	Contents
Academic Reports	Performance, progress, and attendance across classes and subjects.
Examination Reports	Results, grades, and subject-wise analysis.
Financial Reports	Fee collection, dues, and transaction summaries.
Attendance Reports	Daily, monthly, and consolidated attendance.
Library Reports	Books issued, returned, dues, and inventory.
Transport Reports	Route-wise and vehicle utilisation.
GPS & Safety Reports	Trip logs, geofence events, SOS incidents, route-deviation summary.
Parent Engagement Report	Live-view usage, notification response, adoption analytics.
HR & Payroll Reports	Attendance, leave, payroll, and appraisal summaries.

8. Technology Stack & Database

8.1 Technology Stack

Layer	Technology
Frontend (Web)	React.js, TypeScript, Redux Toolkit, Tailwind CSS
Mobile Apps	React Native — Parent, Student, and Driver apps
Backend	Node.js, Express.js (REST + WebSocket)
Database	PostgreSQL with PostGIS extension
Cache & Streams	Redis (also buffers live location events)
Real-Time Layer	WebSockets / MQTT for live GPS ingestion
Maps & Geocoding	Google Maps Platform (Maps SDK, Geofencing, Directions API)
Storage	AWS S3 / S3-compatible object storage
Authentication	JWT + OAuth 2.0 with refresh-token rotation
DevOps	Docker, Kubernetes, GitHub Actions
Monitoring	Prometheus, Grafana, Sentry for error tracking

8.2 Database Schema — Key Entities

Entity	Key Fields
Students	student_id (PK), name, email, phone, class_id

Entity	Key Fields
Transport_Routes	route_id (PK), vehicle_id, driver_id, start_stop, end_stop
Live_Location	location_id (PK), student/vehicle FK, latitude, longitude, timestamp, speed
Geofence_Events	event_id (PK), student_id FK, type (entry/exit), location, timestamp
Attendance	attendance_id (PK), student_id FK, date, status, source
SOS_Alerts	alert_id (PK), student/vehicle FK, location, status, triggered_at
Fees	fee_id (PK), student_id FK, head, amount, due_date, status
Users & Roles	user_id (PK), role, permissions (RBAC), status

9. API Architecture & Security

9.1 Key GPS API Endpoints

Method & Endpoint	Description
POST /api/v1/tracking/location/update	Device pushes live GPS coordinates
GET /api/v1/tracking/student/{id}/live	Parent app fetches live location
POST /api/v1/tracking/sos	Trigger emergency SOS alert
GET /api/v1/tracking/student/{id}/history	Retrieve 30-day trip history
POST /api/v1/tracking/geofence/event	Ingest geofence entry/exit

9.2 Security Requirements

- Role-based access control (RBAC) across all modules.
- JWT / OAuth 2.0 authentication with refresh-token rotation.
- SSL/TLS in transit, AES-256 encryption at rest.
- Password hashing (bcrypt) and strict input validation.
- Protection against OWASP Top 10 (SQL injection, XSS, CSRF, SSRF, IDOR).
- GPS data access strictly limited to verified guardians with full audit logging.
- Configurable location retention policy (default 30 days).
- Regular backups and documented disaster-recovery plan.

10. AI Point-Wise Gap Analysis (Missing / To Add)

Based on a panel-by-panel review of the current PRD, the following items are recommended for inclusion to make the product client-ready and competitive. Each gap is mapped to the affected panel(s) and priority.

Gap / Recommendation	Affected Panel(s)	Priority
Non-Functional Requirements not defined	All Panels	High
Multi-tenant / multi-branch model missing	Admin, Administration, Reports	High
Notification framework under-specified	Comms, Parent, Student, Teacher	High
Parent onboarding & guardian mapping	Parent App, Administration	High
Assignments & LMS-lite features	Teacher, Student, Parent	Medium
Timetable & substitution engine	Academic, Teacher, Admin	Medium
Inventory & Asset Management detail	Admin, Accountant, HR	Medium
Payments — GST, receipts, reconciliation	Accountant, Admin	High
Audit log & data-change history	Admin, All Panels	High
Offline mode for Driver & Teacher apps	Driver App, Teacher	Medium
Data privacy & compliance posture	Admin, Legal	High
Backup, DR, and business continuity	Admin, DevOps	High
Third-party integrations catalogue	Admin, System Settings	Medium
Analytics & AI-assisted insights	Admin, Teacher, Administration	Medium
Accessibility & localisation	All Panels	Medium
Testing, QA, and acceptance criteria	Program Level	High

Detailed Notes on Each Gap

Non-Functional Requirements not defined — [High]

Add explicit targets for performance, availability, scalability, and browser support.

- Target 99.9% uptime SLA and <2s p95 page load.
- Support 5,000 concurrent users and 500 live-tracked buses per tenant.
- Horizontal scaling strategy for WebSocket/MQTT layer.
- Supported browsers, min. Android/iOS versions for mobile apps.

Multi-tenant / multi-branch model missing — [High]

Institutions with multiple branches need tenant isolation and consolidated reporting.

- Tenant-aware data model and RLS-style isolation.

- Branch-level dashboards with roll-up to head office.
- Cross-branch student transfer workflow.

Notification framework under-specified — [High]

SMS/Email/Push are mentioned but the routing, templates and preferences are not.

- Central notification service with per-user channel preferences.
- Template management (with variables) and multilingual support.
- Delivery retries, DLR/bounce tracking, and quiet-hours.
- Emergency channel override for SOS events.

Parent onboarding & guardian mapping — [High]

Multiple guardians per student, custody rules, and app invitation flow are not defined.

- Primary and secondary guardian records with consent flags.
- Invite via SMS/email with OTP-based verification.
- Restricted-access mode for non-custodial guardians.

Assignments & LMS-lite features — [Medium]

Homework and learning material handling are only briefly mentioned.

- Assignment creation, submission window, and late-submission rules.
- File upload with virus scan and size limits.
- Rubric-based evaluation and feedback.
- Plagiarism-check hook (future integration).

Timetable & substitution engine — [Medium]

Class scheduling and teacher-absence substitution are not detailed.

- Auto-generated timetables with clash detection.
- Substitute allocation when a teacher is absent.
- Room and lab allocation.

Inventory & Asset Management detail — [Medium]

Listed as a module but scope is undefined.

- Asset register with barcode/QR tagging and depreciation.
- Stationery, uniform, and lab consumables stock.
- Purchase requisition and vendor management.

Payments — GST, receipts, reconciliation — [High]

Online payment gateway is mentioned; billing controls are not.

- GST-compliant invoices and downloadable receipts.
- Gateway reconciliation report and failed-payment retry.
- Multi-gateway support (Razorpay/Stripe/PayU).

Audit log & data-change history — [High]

Audit is mentioned briefly; needs an application-wide framework.

- Append-only audit log for every write across modules.

- Actor, entity, before/after, IP, and timestamp captured.
- Exportable for compliance review.

Offline mode for Driver & Teacher apps — [Medium]

Rural routes and classrooms often lose connectivity.

- Local queue for attendance and location events.
- Auto-sync on reconnect with conflict resolution.

Data privacy & compliance posture — [High]

Consent is covered for GPS; broader data-protection stance missing.

- Compliance mapping to DPDP Act / GDPR-style principles.
- Data-subject rights: access, correction, erasure workflow.
- Data-processing agreement template for institutions.

Backup, DR, and business continuity — [High]

Backups are mentioned; RPO/RTO targets and drills are not.

- Daily automated backups with off-region copy.
- $RPO \leq 1$ hour, $RTO \leq 4$ hours documented.
- Quarterly DR drill and restore verification.

Third-party integrations catalogue — [Medium]

SMS/Email/Payment/Biometric integrations are implied but not catalogued.

- Plug-and-play connectors with credential vault.
- Health check and failure alerting per integration.

Analytics & AI-assisted insights — [Medium]

Only static reports are covered.

- At-risk student prediction (attendance + grade trend).
- Fee-default risk scoring.
- Route-optimisation suggestions from GPS history.

Accessibility & localisation — [Medium]

Not covered in the current PRD.

- WCAG 2.1 AA compliance for web panels.
- Multi-language UI (English + regional languages).
- RTL support where required.

Testing, QA, and acceptance criteria — [High]

Roadmap mentions UAT; formal QA plan is missing.

- Unit, integration, and end-to-end test coverage targets.
- Load test plan for GPS ingestion (peak school hours).
- Security testing (SAST, DAST, pen-test) before go-live.

11. Delivery Roadmap

The programme is planned in four phases. Timelines will be finalised jointly with the client during kick-off.

Phase	Scope	Key Deliverables
Phase 1	Discovery & Foundation	Requirement finalisation, system design, core module scaffolding.
Phase 2	Build & Integrate	Module implementation, GPS device integration, geofencing engine, Parent
Phase 3	Stabilise & UAT	User acceptance testing including a live GPS pilot with selected buses, bug
Phase 4	Go-Live & Support	Deployment, training, documentation, hyper-care and ongoing support.

12. Appendix

A. Glossary

Term	Meaning
ERP	Enterprise Resource Planning
RBAC	Role-Based Access Control
JWT	JSON Web Token
GPA	Grade Point Average
SOS	Emergency Distress Signal
ETA	Estimated Time of Arrival
MQTT	Lightweight publish/subscribe messaging protocol
PostGIS	Geospatial extension for PostgreSQL

B. Abbreviations

Abbr.	Full Form
PRD	Product Requirements Document
UI	User Interface
DB	Database
API	Application Programming Interface
GPS	Global Positioning System
SLA	Service Level Agreement
RPO/RTO	Recovery Point / Time Objective

C. References

- Internal discussions and stakeholder meetings.
- Industry best practices and standards.
- Client requirement addendum for Live GPS Tracking.

End of Document — Prepared by Mohammad Faizan, EduSphere Solutions.